เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1 — Year: 2018

Subject: 392151 Chemistry I — Section: 15-18

Date: 3 October 2018 — Time: 8.00-10.00

Name: ______________________ ID: __________________ Section: ____________

Instructions:

1. The examination has 8 pages (including this page), 2 sections and a total score of 52 points.
2. Write all your solution and answer on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are **NOT** allowed in the examination.
9. Electronic communication devices are **NOT** allowed in the examination room.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

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Periodic Table of the Elements

1	2		3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
hydrogen 1 H 1.0079																		helium 2 He 4.0026
lithium 3 Li 6.941	beryllium 4 Be 9.0122												boron 5 B 10.811	carbon 6 C 12.011	nitrogen 7 N 14.007	oxygen 8 O 15.999	fluorine 9 F 18.998	neon 10 Ne 20.180
sodium 11 Na 22.990	magnesium 12 Mg 24.305												aluminium 13 Al 26.982	silicon 14 Si 28.086	phosphorus 15 P 30.974	sulfur 16 S 32.065	chlorine 17 Cl 35.453	argon 18 Ar 39.948
potassium 19 K 39.098	calcium 20 Ca 40.078		scandium 21 Sc 44.956	titanium 22 Ti 47.867	vanadium 23 V 50.942	chromium 24 Cr 51.996	manganese 25 Mn 54.938	iron 26 Fe 55.845	cobalt 27 Co 58.933	nickel 28 Ni 58.693	copper 29 Cu 63.546	zinc 30 Zn 65.39	gallium 31 Ga 69.723	germanium 32 Ge 72.61	arsenic 33 As 74.922	selenium 34 Se 78.96	bromine 35 Br 79.904	krypton 36 Kr 83.80
rubidium 37 Rb 85.468	strontium 38 Sr 87.62		yttrium 39 Y 88.906	zirconium 40 Zr 91.224	niobium 41 Nb 92.906	molybdenum 42 Mo 95.94	technetium 43 Tc [98]	ruthenium 44 Ru 101.07	rhodium 45 Rh 102.91	palladium 46 Pd 106.42	silver 47 Ag 107.87	cadmium 48 Cd 112.41	indium 49 In 114.82	tin 50 Sn 118.71	antimony 51 Sb 121.76	tellurium 52 Te 127.60	iodine 53 I 126.90	xenon 54 Xe 131.29
caesium 55 Cs 132.91	barium 56 Ba 137.33	57-70 ■	lutetium 71 Lu 174.97	hafnium 72 Hf 178.49	tantalum 73 Ta 180.95	tungsten 74 W 183.84	rhenium 75 Re 186.21	osmium 76 Os 190.23	iridium 77 Ir 192.22	platinum 78 Pt 195.08	gold 79 Au 196.97	mercury 80 Hg 200.59	thallium 81 Tl 204.38	lead 82 Pb 207.2	bismuth 83 Bi 208.98	polonium 84 Po [209]	astatine 85 At [210]	radon 86 Rn [222]
francium 87 Fr [223]	radium 88 Ra [226]	89-102 ■	lawrencium 103 Lr [262]	rutherfordium 104 Rf [261]	dubnium 105 Db [262]	seaborgium 106 Sg [266]	bohrium 107 Bh [264]	hassium 108 Hs [269]	meitnerium 109 Mt [268]	darmstadtium 110 Ds [271]	roentgenium 111 Rg [272]	ununbium 112 Uub [277]	ununtrium 113 Uut [284]	ununquadium 114 Uuq [289]	ununpentium 115 Uup [288]	ununhexium 116 Uuh [292]	ununseptium 117 Uus [291]	ununoctium 118 Uuo [294]

■Lanthanide series	lanthanum 57 La 138.91	cerium 58 Ce 140.12	praseodymium 59 Pr 140.91	neodymium 60 Nd 144.24	promethium 61 Pm [145]	samarium 62 Sm 150.36	europium 63 Eu 151.96	gadolinium 64 Gd 157.25	terbium 65 Tb 158.93	dysprosium 66 Dy 162.50	holmium 67 Ho 164.93	erbium 68 Er 167.26	thulium 69 Tm 168.93	ytterbium 70 Yb 173.04
■Actinide series	actinium 89 Ac [227]	thorium 90 Th 232.04	protactinium 91 Pa 231.04	uranium 92 U 238.03	neptunium 93 Np [237]	plutonium 94 Pu [244]	americium 95 Am [243]	curium 96 Cm [247]	berkelium 97 Bk [247]	californium 98 Cf [251]	einsteinium 99 Es [252]	fermium 100 Fm [257]	mendelevium 101 Md [258]	nobelium 102 No [259]

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Given : Speed of light; c = 3.0×10^8 m/s Plank's constant; h= 6.6×10^{-34} J.s

	Multiple Choice	Part 2(1)	Part 2(2)	Part 2(3)	Total
score					

Part 1. Multiple Choice (17 points)

Choose the letter of the choice in the answer table that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				

Question no.	a.	b.	c.	d.
9				
10				
11				
12				
13				
14				
15				
16				
17				

1. Orange juice with pulp is an example of a(n)________, while syrup is an example of a(n)________.

   a. homogeneous mixture, pure substance b. element, compound

   c. heterogeneous mixture, homogeneous mixture c. solution, alloy

2. An element can be defined as ____________________________________.

   a. a substance that cannot be separated into two or more substances by ordinary chemical (or physical) means.

   b. a substance with constant composition

   c. a substance that contains two or more substances, in definite proportion by weight

   d. a uniform mixture

3. Which set of chemical name and chemical formula for the same compound is correct?

   a. magnesium dichromate, $MgCrO_4$ b. Iron(III) phosphate, $FePO_4$

   c. ammonium sulfite, $(NH_4)_2S$ d. lithium carbonate, $LiCO_3$

4. Which polyatomic ion forms a neutral compound when combined with a group 1A monatomic ion in a 1:1 ratio?

   a. ammonium b. nitrate

c. carbonate d. phosphate

5. Which of the following is <u>not</u> the properties of metal?

a. Electrical conductor b. Shiny metallic lustre

c. Malleable d. Thermal insulator

6. Which of the following could indicate that a chemical change has taken place?

a. evaporation b. distillation c. synthesis d. melting

7. How many protons, neutrons and electrons are there in an atom of aluminium (Al)?

	protons	neutrons	electrons
a.	10	27	13
b.	13	14	13
c.	13	14	27
d.	27	13	14

8. In Thomson's plum-pudding model of the atom, the plums represent __________ .

a. protons b. electrons c. neutrons d. atoms

9. In _____ atomic model, negative electrons orbit the positively charged nucleus.

a. Dalton's b. Thomson's c. Rutherford's d. Democritus's

10. According to Bohr's theory, an electron's path around the nucleus defines its __________ .

a. electric charge b. energy level c. atomic mass d. speed

11. If electrons in an atom have the lowest energies, the electrons are in their _____________ .

a. ground states b. excited states

c. inert states d. radiation-emitting states

12. Atoms of ^{16}O, ^{17}O, and ^{18}O have the same number of_____________ .

a. protons, but a different number of electrons b. electrons, but a different number of protons

c. protons, but a different number of neutrons d. neutrons, but a different number of protons

13. Green light has a frequency of 6.0×10^{14} Hz. What is the wavelength?

a. 400 nm b. 500 nm c. 600 nm d. 700 nm

14. Which one of the following statements is incorrect?

a. s orbitals are found in all principal energy levels.

b. s orbitals can only hold one electron.

c. Each p orbital can hold up to two electrons.

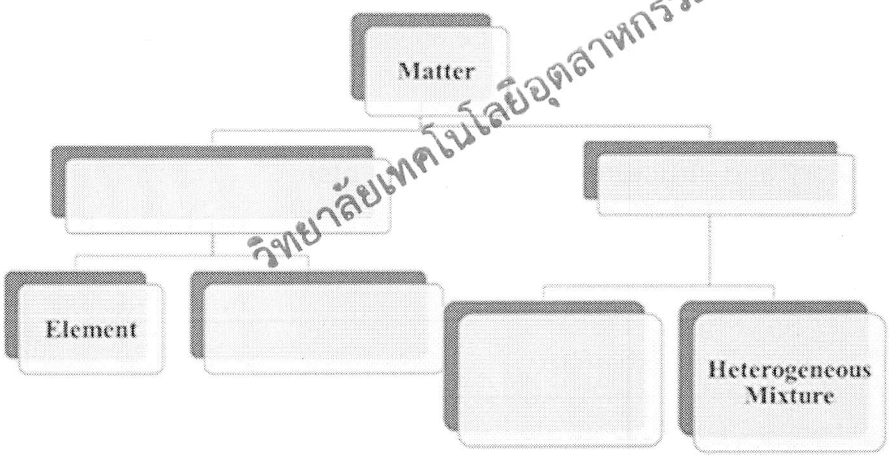
d. d orbitals are capable of holding less than eleven electrons.

15. Why should you fill electrons in the 4s sub-level before the 3d sub-level?
    a. There is only one 4s orbital and there are 5 3d orbitals.
    b. The 4s orbital is spherical.
    c. The 4s orbital has an energy lower than the 3d orbital.
    d. The 3d orbitals have a lower energy.
16. The electronic configuration of an element with atomic number 8 is
    a. $1s^2 2s^6$ b. $1s^2 2s^2 2p^6$ c. $2s^2 2p^6$ d. $1s^2 2s^2 2p^4$
17. Which one of the following is not the electronic configuration of a noble gas atom?
    a. $1s^2$ b. $1s^2 2s^2$ c. $1s^2 2s^2 2p^6$ d. $1s^2 2s^2 2p^6 3s^2 3p^6$

Part 2 Write your answer in the space provided for each question (35 points)

1. Matter Classification, Nomenclature, Properties, Physical and Chemical Changes. (9 pts)

1.1 Fill proper terms in 4 boxes of the matter classification flow chart as shown below. (2 pts)



1.2 Complete the following sentences by the appropriate word(s) from the list below. Each word can be used once. (2 pts)

filtration	magnetic separation	evaporation	distillation
heating	chromatography	centrifugation	crystallization

1.2.1 The separation technique that takes advantage of different boiling points is called ____________________.

1.2.2 Separating sand from water can be done by ____________________.

1.2.3 The water in sugar water can be removed by ____________________.

1.2.4 Sulfur and magnetic particles can be separated by ____________________.

1.3 Write the chemical name or the chemical formula for each of the following. (2 pts)

1.3.1 $NaNO_3$..

1.3.2 N_2O_5..

1.3.3 Calcium sulfate..

1.3.4 Aluminium iodide..

1.4 Specify the following sentences whether are TRUE or FALSE. If any sentence is false, you must correct it. (3 pts)

1.4.1 Flammability is physical property.

....

1.4.2 Boiling point and melting point are not chemical property.

....

1.4.3 Metal can conduct only heat.

....

1.4.4 Plastic is usually used to cover the copper wire since it is light weight.

....

1.4.5 Rust formation is one of the chemical changes.

....

1.4.6 Lighting fireworks is claimed to be an irreversible reaction.

....

2. Atomic theory and structure of the atom. (9 pts)

2.1 Choose words from the list as shown below to fill in the blanks in the paragraphs. (3 pts)

J.J. Thompson	Avogadro	Millikan	Bohr
Dalton	Rutherford	Gay-Lussac	Aufbau
Hund (degeneracy)	Pauli (Exclusive)	atomic orbital	Planck
Louise de Broglie	Heisenberg		

a. ____________ used the Cathode Ray Tube to discover the electron and determined its charge to mass ratio.

b. “A region of space in which there is a high probability of finding an electron” is the definition of__________________ .

c. The scientist whose alpha-particle scattering experiment led him to conclude that the nucleus of an atom contains a dense center of positive charge is_____________________ .

d. The planetary model of the atom was stated by _____________________.

f. The oil drop experiment was performed by ___________________ to measure the elementary electric charge (the charge of the electron).

g. The rule that electrons will enter empty orbitals of a sublevel before they pair up__________.

2.2 Use the periodic table to fill in the following chart. (3 pts)

Element	Symbol	Atomic Number	Atomic Mass	Number of protons	Number of electrons	Number of Neutrons	Period	Valence
Silicon								
	${}^{14}_{6}C$							
Iodine-131								

2.3 A helium laser emits light with a wavelength of 633 nm. What is the frequency and energy of the light? (3 pts)

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3. Electron configuration and Periodic table. (17 pts)

3.1 Write the electron configurations for the following ions and elements. (6 pts)

Element, Ion	Electron configuration (using the subshell model)	Electron configuration (using the shell model)
Br		
Cu		
Ni^{2+}		

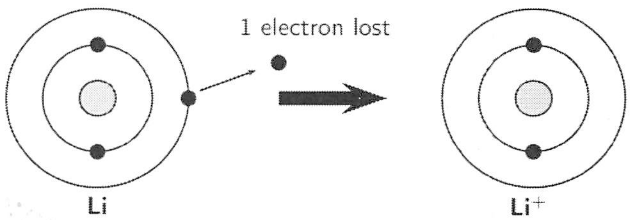
3.2 Write the full electron configuration and fill in the orbital diagrams, for the following elements. (4 pts)

Here is an example of the hydrogen : $1s^2$ ____________________

↿
1S

Chlorine:	____
Chromium:	____

3.3 Use the diagram for lithium as a guide and draw similar diagrams to show how each of the following ions is formed. (4 pts)



Mg^{2+}	
O^{2-}	

3.4 Determine what group (1-18) and period of the elements which are denoted by the following electron configurations (3 pts)

	group (1-18)	period
$1s^22s^22p^63s^23p^4$		
$1s^22s^22p^63s^23p^64s^23d^{10}4p^65s^1$		
[Kr] $5s^24d^{10}5p^3$		

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